MATH 189 - Data Analysis

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Contents

1 Statistical Learning

1.1 Terminology

"Supervised Learning" means we have both predictors and responses.
"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, ..., X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$
We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of X} \end{cases}$$

Question: Why estimate f?
Prediction: Given X, we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned} \mathbb{E}(y - \hat{y})^2 &= \mathbb{E}(f(x) + \epsilon - \hat{f}(x)) \\ &= \mathbb{E}(f(x) - \hat{f}(x))^2 + 2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2) \end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x) - \hat{f}(x))\epsilon) = \mathbb{E}(f(x) - \hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y - \hat{y})^2 = (f(x) - \hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

"Inference" - Sometimes we want to understand the association between y and $X_1, ..., X_p$

1. Which predictors are associated with response?
2. Is the association positive or negative?
3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$(x_1, y_1), (x_2, y_2), ...(x_n, y_n)$$
$$\text{where } X_i = (X_{i1}, X_{i2}, ..., X_{ip})$$

n = # data points (e.g. $n = 30$)
p = # predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

Defined on training data

$$\text{Test MSE} = \text{ Ave } (y_0 - \hat{f}(x_0))^2$$

Defined on previously unseen data point (x_0, y_0)

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p$$

"Fit/train the model" usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, ..., \beta_p$ that make the MSE and hopefully the test MSE small.

Classification
Data set $(x_1, y_1), (x_2, y_2,), ..., (x_n, y_n)$ where $X_i = (x_{i1}, ..., x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}\{y_i \neq \hat{y}_i\}$$

$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)

The KNN classifier is defined by letting η_0 = set of indices $i \in \{1, ..., n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j.

2 Linear Regression

2.1 Simple Linear Regression

Underlying statistical model: $y = \beta_0 + \beta_1 X + \epsilon$ where y is the quantitative response and a single predictor X which is why it is called simple.

Prediction model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
$\hat{\beta}_0, \hat{\beta}_1$ are estimates based on training data, where hat means estimated quantity.

<u>Estimating coefficients</u>
The residual sum of squares (RSS)

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \end{aligned}$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS.

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \hat{\beta}_0} &= \frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 X)^2 \\ &= 2 \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ &\implies \text{RSS is } \begin{cases} \text{decreasing for really small } \hat{\beta}_0 \\ \text{increasing for really big } \hat{\beta}_0 \end{cases} \end{aligned}$$

RSS is optimal (in $\hat{\beta}_0$) when

$$\begin{aligned} 0 &= \frac{1}{n}\sum_{i=1}^{n}(\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ &= \hat{\beta}_0 + \hat{\beta}_1\Big(\frac{1}{n}\sum_{i=1}^{n} X_i\Big) - \Big(\frac{1}{n}\sum_{i=1}^{n} Y_i\Big) \\ &= \hat{\beta}_0 + \hat{\beta}\bar{x} - \bar{y} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1\bar{x} \end{aligned}$$

Where $\begin{cases} \bar{x} = \frac{1}{n}\sum_{i=1}^{n} X_i \text{ is the mean predictor} \\ \bar{y} = \frac{1}{n}\sum_{i=1}^{n} Y_i \text{ is the mean response} \end{cases}$

Covariance

$$\begin{aligned} \operatorname{Cov}(X_i, Y_i)_{i=1}^{n} &= \frac{1}{n}\sum_{i=1}^{n} X_i Y_i - \bar{X}\bar{Y} \\ &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})(Y_i - \bar{y}) \end{aligned}$$

Variance

$$\begin{aligned} \operatorname{Var}(X_i)_{i=1}^{n} &= \operatorname{Cov}(X_i, X_i)_{i=1}^{n} \\ &= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})^2 \end{aligned}$$

From this we can get our optimal $\hat{\beta}_1$

$$\begin{aligned} \hat{\beta}_1 &= \frac{\operatorname{Cov}(X_i, Y_i)_{i=1}^{n}}{\operatorname{Var}(X_i)_{i=1}^{n}} \\ &= \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})(Y_i - \bar{y})}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})^2} \end{aligned}$$

2.2 Goodness of Fit & Correlation

Assume an underlying statistical model

$$y = \beta_0 + \beta_1 X + \epsilon$$

We use training data $(X_i, Y_i)_{i=1}^n$ to generate <u>prediction model</u> $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize RSS:

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 = \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \end{cases}$$

<u>Goodness of fit</u>

$$\begin{aligned}
\text{RSS} &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\
&= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\
&= \sum_{i=1}^{n} \Big((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \Big)^2 \\
&= \sum_{i=1}^{n} (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2 \sum_{i=1}^{n} (x_i - \bar{x})^2 \\
&= \sum_{i=1}^{n} (y_i - \bar{y})^2 - \frac{\Big[\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y}) \Big]^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}
\end{aligned}$$

RSS scales with the number of data points n and also scales with y^2 units.
$\rightarrow$ We can fix the scaling and get out a goodness-of-fit score (R^2) between 0 and 1.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$\text{RSS} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \qquad\qquad \text{TSS} = \sum_{i=1}^{n} (y_i - \bar{y})^2$$

So for simple linear regression,

$$
\begin{aligned}
R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} \\
&= \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \\
&= \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} \\
R &= \frac{\text{Cov}((x_i), (y_i))}{\text{Var}(x_i)^{1/2}\text{Var}(y_i)^{1/2}}
\end{aligned}
$$

Long story short R^2 is the fraction of variance explained by the model.

$\boxed{ex}$ $\quad \hat{\text{sales}} = 7.0 + 0.048 \cdot \text{TV}$
$R^2 = 0.61 = 61\%$ $\qquad r \approx 0.78$ pretty strong positive correlation

2.3 Regression Table

$\boxed{ex}$

	Coefficient	Std. error	t-statistic	p-value
Intercept (β_0)	7.0	0.46	15	<0.0001
TV (β_1)	0.048	0.0027	18	<0.0001

Std error is the error bar on linear coefficients.

$$
\text{We know that } \begin{cases} \beta_0 \approx \hat{\beta}_0 \pm \text{SE}(\hat{\beta}_0) \\ \beta_1 \approx \hat{\beta}_1 \pm \text{SE}(\hat{\beta}_1) \end{cases}
$$

Assume the underlying statistical model is true!

$$
y_i = \beta_0 + \beta_1 X_i + \epsilon_i
$$
$$
\text{where } \epsilon_i \text{ are independent } \eta(0, \sigma^2) \text{ random variables}
$$

If we generate N data points (x_i, y_i) from the model, then

$$
\begin{cases} \text{Var}(\hat{\beta}_0)^{1/2} \approx \text{Std.}\hat{\ }\text{error}(\hat{\beta}_0) \\ \text{Var}(\hat{\beta}_1)^{1/2} \approx \text{Std.}\hat{\ }\text{error}(\hat{\beta}_1) \\ \mathbb{E}(\hat{\beta}_0) = \beta_0 \\ \mathbb{E}(\hat{\beta}_1) = \beta_1 \end{cases}
$$

2.4 Multiple Linear Regression

The underlying statistical model that we assume is true.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + ... + \beta_p x_p$$

where y is response, β_0 is the intercept, $\beta_1 x_1, ..., \beta_p x_p$ are predictors, and ϵ is the mean zero noise.

$\boxed{ex}$ sales $= \beta_0 + \beta_1 \text{TV} + \beta_2 \text{Radio} + \beta_3 \text{Newspaper} + \epsilon$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + ... + \hat{\beta}_p x_p$$

We want to find the best $\hat{\beta}$ to minimize RSS, where $\hat{\beta}$ are estimated quantities.

$$\begin{aligned} RSS &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta} x_{ip})^2 \\ &= \left|\left| \begin{bmatrix} y_1 - \hat{\beta}_0 - \hat{\beta}_1 x_{i1} - ... - \hat{\beta}_p x_{ip} \\ \vdots \\ y_n - \hat{\beta}_0 - \hat{\beta}_1 x_{n1} - ... - \hat{\beta}_p x_{ip} \end{bmatrix} \right|\right|^2 \\ &= ||Y - X\hat{\beta}||^2 \end{aligned}$$

where $y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \qquad \hat{\beta} = \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_p \end{bmatrix} \qquad X = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ 1 & x_{21} & \dots & x_{1p} \\ & & \ddots & \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}$

The best $\hat{\beta}$ to minimize Euclidean norm^2 for $||Y - X\hat{\beta}||^2$ is $\hat{\beta} = X^+ Y = (X^T X)^{-1} X^T Y$

	Coefficient	Std. error	t-statistic	p-value
Intercept	$\hat{\beta}_0$ 2.939	0.3119	9.42	<0.0001
TV	$\hat{\beta}_1$ 0.046	0.0014	32.81	<0.0001
Radio	$\hat{\beta}_2$ 0.189	0.0086	32.89	<0.0001
Newspaper	$\hat{\beta}_3$ -0.001	0.0059	-0.18	0.8599

Note: the standard error is NOT $Var(\hat{\beta}_i)^2$ but rather $\hat{Var}(\hat{\beta}_i)^2$

Hypothesis testing

State a hypothesis, test it with data (testing the likelihood such data could possibly be generated if the hypothesis were true).

H_0 - the null hypothesis: $y = \beta_0+\beta_1 x_1+\beta_2 x_2+\beta_3 x_3+\epsilon$ for some coefficients B_0, β_2, β_3 with $\beta_1 = 0$ with $\epsilon \sim N(0, \sigma^2)$ for some variance $\sigma^2 \geq 0$. Also, observations are independent.

H_1 - the alternative hypothesis: $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$ where everything above holds except $\beta_1 \neq 0$

Assume H_0. Then

$$\frac{\hat{\beta}_1}{\hat{SE}(\hat{\beta}_1)} = t \sim T_{n-p-1}$$

3 Logistic Regression & Generative Models

3.1 Logistic Regression

We want to predict whether an event happens ($y = 1$) or not ($y = 0$), we really want to model the probability $p \in (0, 1)$ of occurrence $\implies$ NOT LINEAR REGRESSION.

Before

$$y = \beta_0 + \beta_1 X_\epsilon$$

Now

$$p = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$
$$1 - p = \frac{1}{1 + \exp(\beta_0 + \beta_1 X)}$$
$$\frac{p}{1-p} = \exp(\beta_0 + \beta_1 X)$$

Where the term $\frac{p}{1-p}$ is called "odds".
Note: $\log(\text{odds}) = \beta_0 + \beta_1 X$

$\boxed{ex}$ Defaulting on bank loans

$$\begin{aligned} p &= \text{probability of defaulting} \\ X &= \text{balance} \\ y_i &= \begin{cases} 0 & \text{default doesn't happen} \\ 1 & \text{default does happen} \end{cases} \\ \log(\frac{p}{1-p}) &= \beta_0 + \beta_1 \cdot \mathrm{X} \end{aligned}$$

How do we use data/observations $(x_i, y_i)_{1\le i\le n}$ to bestimate β_0, β_1.

$$\begin{cases} \text{Linear regression} \implies \text{minimize RSS} \\ \text{Logistic regression} \implies \text{use max likelihood, do it on a computer} \\ \text{Generative models} \implies \text{simple formulas for } \hat{\Pi}_k, \hat{\mu}_k, \hat{\Sigma}_k \end{cases}$$

We choose estimates $\hat{\beta}_0, \hat{\beta}_1$ to maximize likelihood of observations $(x_i, y_i)_{1\le i\le n}$ if (a) the observations were independent and (b) they were generated exactly from the statistical model.

$$\begin{aligned} \mathbb{P}\left((x_i, y_i)_{1\le i\le n}\right) &= \prod_{i=1}^{n} \mathbb{P}(x_i, y_i) \text{ where } y_i \in \{0, 1\} \\ &= \prod_{y_i=1} \hat{p}(x_i) \prod_{y_i=0} (1 - \hat{p}(x_i)) \\ &= \prod_{i:y_i=1} \frac{\exp(\beta_0 + \beta_1 X_i)}{1 + \exp(\beta_0 + \beta_1 X_i)} \prod_{i:y_i=0} \frac{1}{1 + \exp(\beta_0 + \beta_1 X_i)} \end{aligned}$$

Know that $\hat{\beta}_0, \hat{\beta}_1$ are chosen to maximize this probability/likelihood above.

$\boxed{ex}$ $P_{\text{defaulting}}$

	Coefficient	Std. error	z-statistic	p-value
Intercept	-10.6513	0.3612	-29.5	<0.0001
Balance	.0055	0.0002	24.9	<0.0001

We want $\hat{\beta}_0 + \hat{\beta}_1 X = 0$

$$
\begin{aligned}
-10.6513 + 0.0055X &= 0 \\
X &= \frac{10.6513}{0.0055} \\
&= 1936.6
\end{aligned}
$$

Assume

a. The model is true

b. Observations $(x_i, y_i)_{1 \le i \le n}$ are independent draws from the model

c. Model is true with $\beta_0 = 0$ (null hypothesis)

3.2 Generative Models for Classification

Predictors
X or $X_1, X_2, ..., X_n$
Response
$y \in \{0,1\}$ or $y \in \{1, 2, ..., k\}$

<u>Review</u>
Simple logistic regression

$$P = \text{prob}\{y = 1\} = \frac{\exp(\beta_0 + \beta_1 X)}{1 + \exp(\beta_0 + \beta_1 X)}$$

Note: This doesn't make any assumptions about how predictors are distributed.

For each $k = 1, 2, ..., K$:
Our statistical model ('mixture of Gaussians') says $y = k$ with probability π_k $(\sum_{k=1}^{K} \pi_k = 1)$
Given $y = k$, $X \sim N(\mu_k, \Sigma_k)$ where Σ is the covariance matrix symbol. Here is the general vector-valued format:

$$X = (X_1, ..., X - p) \qquad \Sigma_k = \begin{bmatrix} \quad \end{bmatrix} \text{pxp matrix}$$

Where μ_k has Kp parameters and Σ_k has $K\frac{p(p+1)}{2}$ parameters
When $p = 1$, we can write

$$X \sim N(\mu_k, \sigma_k^2)$$

In approaches like Linear Discriminant Analysis (LDA), Quadratic Discriminant Analysis (QDA), and Naive Bayes (NB) use simple estimators for Π_k, μ_k, Σ_k that we can calculate fast.

$\boxed{ex}$ Linear discriminant analysis (LDA) make assumptions of common covariance structure $\Sigma = \Sigma_1 = \Sigma_2 = ... = \Sigma_k$ which reduces a lot of complexity. Given observations $(X_i, y_i)_{1 \leq i \leq n}$, we can estimate for $k = 1, ..., K$

$$\begin{aligned}
\hat{\Sigma} &= \frac{n_k}{n}, \text{ where} \\
n &= \#\{i : y_i = k\} \\
&= \text{number of observations in class } k \\
\hat{\mu}_k &= \frac{1}{n_k} \sum_{i:y_i=k} X_i
\end{aligned}$$

Last estimate common covariance by

$$\begin{aligned}
\hat{\Sigma} = \frac{1}{n-k} \sum_{k=1}^{K} \sum_{i:y_i=k} (X_i - \mu_k)(X_i - \mu_k)^T \\
\mathbb{E}\hat{\Sigma} = \Sigma \text{ if the model is true}
\end{aligned}$$

$\boxed{ex}$ Quadratic discriminant analysis is pretty much the same except DIFFERENT covariances $\Sigma_1, ..., \Sigma_k$ for each class.
$\implies \hat{\Pi}_k = \frac{n_k}{n}$, $\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} X_i$
$\implies \hat{\Sigma}_k = \frac{1}{n_k - 1} \sum_{i:y_i=k} (X_i - \mu_k)(X_i - \mu_k)^T$
$\implies K\frac{p(p+1)}{2}$ covariance free parameters.
$\boxed{ex}$ Naive Bayes is pretty much the same except DIFFERENT covariances $\Sigma_1, ..., \Sigma_k$ that are also diagonal matrices.
$\implies Kp$ covariance free parameters with the same estimators as QDA.

$$\text{diag}\left(\sum_{k}^{n}\right) = \frac{1}{n_k - 1} \sum_{i:y_i=k} (X_i - \mu_k) \odot (X_i - \mu_k)$$

Suppose you have the statistical model with probability Π_k, we set $y = k$ and draw $X \sim N(\mu_k, \Sigma_k)$. Suppose we have estimated the parameters well - see a new predictor vector $X = (X_1, ..., X_p)$ - How do you determine class probabilities $P\{y = k|X\}$ and make predictions?

Prediction part is easy - just predict based on the highest probability class: $\hat{y} = k$ for $k = \text{argmax}_{1 \le j \le k} \; P\{y = j|X\}$

So what is $P\{y = k|X\}$? Bayes' rule tells us

$$\begin{aligned} P\{y = k|X\} &= \frac{P\{x|y = k\} \; P\{y = k\}}{\sum_{j=1}^{k} P\{X|y = j\}P\{y = j\}} \\ &= P(X) \\ &= \frac{f_k(X)\Pi_k}{\sum_{j=1}^{K} f_j(X)\Pi_j} \end{aligned}$$

$$\text{where Gaussian density } \Pi_j(X) = \frac{1}{(2\pi)^{p/2}(\det \Sigma)^{1/2}} \exp(\frac{-1}{2}(X - \mu_j)\Sigma_j(X - \mu_j))$$